LAB 1: Problem Statement: Hostel Maintenance & Issue Ticketing System

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SECTION: G

Problem Number: 7

FUNCTIONAL REQUIREMENTS TABLE

ID	Priority	Requirement Description	Acceptance Criteria	Rationale
FR-001	**High**	**The system shall allow registered hostel residents to create maintenance issue tickets with category, room/location, description, priority, and optional photo or attachment.**	**Pass: A resident submits all required details and receives a unique ticket ID. Fail: The ticket cannot be submitted or no ticket ID is generated.**	**Ensures hostel issues are formally recorded and traceable.**
FR-002	**High**	**The system shall validate ticket information and reject submissions with missing required fields or unauthorized access.**	**Pass: Invalid or unauthorized submissions are rejected. Fail: Invalid tickets enter the maintenance workflow.**	**Prevents incomplete or unauthorized complaints from entering the system.**
FR-003	**High**	**The system shall process tickets through Open, Assigned, In Progress, Resolved, and Closed statuses and record each status change.**	**Pass: A valid ticket starts as Open and every status change is recorded. Fail: Status changes are not recorded.**	**Provides reliable issue tracking and visibility into maintenance progress.**
FR-004	**High**	**The system shall allow wardens/administrators to assign tickets to maintenance staff and allow staff to update progress and resolution details.**	**Pass: Authorized users can assign and update tickets. Fail: Tickets cannot be assigned or updated.**	**Ensures issues reach responsible personnel and can be actively managed.**
FR-005	**Medium**	**The system shall allow authorized users to search and filter tickets by ticket ID, status, category, priority, room, and date.**	**Pass: Users can retrieve relevant tickets using supported filters. Fail: Relevant tickets cannot be located.**	**Helps users monitor complaints and diagnose unresolved issues.**
FR-006	**Medium**	**The system shall generate maintenance reports containing ticket counts by status, category, priority, and resolution information.**	**Pass: Reports accurately display maintenance statistics and details. Fail: Reports contain missing or incorrect information.**	**Supports monitoring, auditing, workload analysis, and planning.**

NON-FUNCTIONAL REQUIREMENTS TABLE

ID	Priority	Requirement Description	Acceptance Criteria	Rationale
NFR-001	High	The system shall support at least 1,000 concurrent ticket-related requests and acknowledge valid submissions within 3 seconds for at least 95% of requests.	Pass: At least 95% complete within 3 seconds during load testing. Fail: More than 5% exceed 3 seconds.	Ensures responsiveness during high hostel usage.
NFR-002	High	The system shall authenticate users, enforce role-based authorization, protect ticket data using secure communication, and maintain at least 99.5% monthly availability.	Pass: Unauthorized access is blocked, data is protected, and availability is at least 99.5%. Fail: Security or availability requirements are not met.	Protects resident information and ensures dependable access.
NFR-003	High	The system shall prevent duplicate ticket creation when the same issue submission is delivered or submitted more than once.	Pass: Duplicate submissions do not create unintended duplicate tickets. Fail: Duplicate requests create multiple tickets.	Prevents duplicate complaints and unnecessary maintenance work.
NFR-004	Medium	The system shall retain ticket history, assignments, status updates, timestamps, resolution details, and relevant user actions for auditing and troubleshooting.	Pass: Authorized administrators can retrieve required history for the retention period. Fail: Required history is missing.	Provides traceability and simplifies investigation of unresolved issues.

USE CASE FLOW SPECIFICATION

Use Case ID: UC-001

Use Case Name: Create and Process Maintenance Issue Ticket

Primary Actor: Hostel Resident, Hostel Administrator/Warden

Secondary Actor: Maintenance Staff, Notification Service

Goal: The resident reports a hostel maintenance issue so that it can be validated, assigned, resolved, and tracked to closure.

Preconditions

1. The resident or authorized user is registered and logged in.
2. Hostel rooms/locations and maintenance categories are configured.
3. The ticketing service is available.
4. Maintenance staff or an administrator is available for assignment.

Postconditions

1. The ticket is stored with a unique ticket ID.
2. The ticket is assigned the initial status Open.
3. The ticket is assigned to appropriate maintenance staff or an administrator.
4. Progress and resolution details are recorded.
5. A resolved ticket can be closed after confirmation or administrative review.
6. The final status and complete activity history are retained.

Main Success Scenario

1. The resident logs into the hostel maintenance system.
2. The resident selects Create New Maintenance Ticket.
3. The system displays the ticket submission form.
4. The resident enters the issue category, room/location, description, priority, and optional attachment.
5. The system validates the required information and user authorization.
6. The system creates the ticket and assigns a unique ticket ID.
7. The system stores the ticket with status Open.
8. The administrator/warden reviews the new ticket.
9. The administrator assigns the ticket to appropriate maintenance staff.
10. The assigned staff member accepts the ticket and changes its status to In Progress.
11. The staff member performs the required maintenance work and records resolution details.
12. The system updates the ticket status to Resolved.
13. The resident or authorized administrator reviews the resolution and the ticket is marked Closed.
14. The system records the final status, timestamps, user actions, and resolution information.

Alternate Flow

AF-01: Invalid Ticket Submission

At Step 5, if required ticket information is missing or invalid:

1. The system identifies the invalid or incomplete information.
2. The system rejects the ticket submission.
3. The system displays an appropriate validation message.
4. The resident corrects the information and resubmits the ticket.
5. The ticket is not processed until validation succeeds.

AF-02: Maintenance Processing Failure

At Step 11, if the maintenance issue cannot be resolved immediately:

1. The staff member records the reason for the delay or failure.
2. The system retains the ticket in In Progress or another configured pending state.
3. The administrator may reassign the ticket or escalate its priority.
4. The system records the reassignment or escalation in the ticket history.
5. The staff member attempts resolution again.
6. If the issue is successfully resolved, the ticket proceeds to Resolved.

AF-03: Duplicate Issue Ticket

At Step 6, if the system detects an existing active ticket for the same resident, room, category, and issue:

1. The system identifies the submission as a possible duplicate.
2. The system does not create an unintended duplicate ticket.
3. The system displays the existing ticket ID and status.
4. The duplicate submission event is recorded where required for auditing.

AF-04: Ticket Reopened

After Step 13, if the resident reports that the issue has not actually been resolved:

1. The system reopens the ticket.
2. The ticket status changes to Reopened.
3. The resident provides additional comments if required.
4. The administrator reassigns or escalates the ticket.
5. The previous resolution and new activity remain in the ticket history.

